

Correspondence coloring graphs with a forbidden clique

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Abstract

For every fixed integer $r \geq 4$, we prove that every K_r -free graph of sufficiently large maximum degree Δ has correspondence chromatic number $O_r(\Delta/\log \Delta)$. This resolves the Alon–Krivelevich–Sudakov coloring conjecture in the stronger correspondence-coloring form. The same bound, with a constant depending on F , holds when any fixed graph F is excluded as an ordinary subgraph. Ordinary and list coloring satisfy the same bounds.

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1 Introduction

All graphs in this paper are finite, simple, and undirected. A *correspondence assignment* on a graph G consists of pairwise disjoint finite sets $L(v)$, one for each vertex, and a matching $M_{uv} \subseteq L(u) \times L(v)$ for each edge uv . A coloring of the assignment chooses $c(v) \in L(v)$ at every vertex so that $(c(u), c(v)) \notin M_{uv}$ for every edge uv . The matchings need not agree with one another around cycles. The *correspondence chromatic number*, also called the *DP-chromatic number*, $\chi_{\text{DP}}(G)$, is the least integer k for which every correspondence assignment with $|L(v)| \geq k$ admits a coloring.

This notion includes list coloring: replace each color in each list by a distinct copy, and match copies of the same color along an edge. Ordinary coloring is obtained by giving every vertex the same list before taking these copies. Thus

$$\chi(G) \leq \chi_{\text{list}}(G) \leq \chi_{\text{DP}}(G) \leq \Delta(G) + 1,$$

where the last inequality follows by greedy coloring. We prove a logarithmic improvement when the clique number is bounded.

Theorem 1.1. *For every integer $r \geq 4$ there are constants $C_r > 0$ and $\Delta_r \geq 3$ such that every K_r -free graph G of maximum degree $\Delta \geq \Delta_r$ satisfies*

$$\chi_{\text{DP}}(G) \leq \left\lceil C_r \frac{\Delta}{\log \Delta} \right\rceil.$$

The constants depend only on r .

In particular, the estimate is uniform in the number of vertices, the sizes of the lists, and the edge matchings. Throughout the paper, all logarithms are natural. We do not optimize the constant or the degree threshold.

For a fixed finite graph F , call G *F-free* if it contains no subgraph isomorphic to F ; the subgraph need not be induced.

Corollary 1.2. *For every finite graph F there are constants $C_F > 0$ and $\Delta_F \geq 3$ such that every F -free graph G of maximum degree $\Delta \geq \Delta_F$ satisfies*

$$\chi_{\text{DP}}(G) \leq \left\lceil C_F \frac{\Delta}{\log \Delta} \right\rceil.$$

Proof. Set $r = \max\{4, |V(F)|\}$. Since K_r contains F as a subgraph, every F -free graph is K_r -free. Apply Theorem 1.1. $\square$

The ordinary-coloring version of Corollary 1.2 is Conjecture 3.1 of Alon, Krivelevich, and Sudakov [2]. The corollary therefore proves that conjecture in the stronger correspondence-coloring form. If F contains a cycle, the order $\Delta/\log \Delta$ cannot be improved in general: graphs with girth larger than $|V(F)|$ can have maximum degree at most Δ and chromatic number of this order [10]; see also the proof of [2, Proposition 1.2].

1.1 Historical context

List coloring was introduced independently by Vizing [26] and by Erdős, Rubin, and Taylor [15]. Dvořák and Postle introduced correspondence coloring in their 2015 preprint, published as [13], to

allow the identification of colors to vary from edge to edge. This flexibility was central to their application to list coloring planar graphs. An arbitrary correspondence assignment need not admit globally consistent color names, so a list-coloring argument that uses independence across such names requires additional work.

An early step toward logarithmic coloring bounds imposed a stronger restriction on short cycles. Kim [19] proved $\chi_{\text{list}}(G) \leq (1 + o(1))\Delta/\log \Delta$ for graphs of girth at least five. Johansson subsequently obtained the list-coloring bounds $O(\Delta/\log \Delta)$ for triangle-free graphs and $O_r(\Delta \log \log \Delta/\log \Delta)$ for K_r -free graphs. His triangle-free argument is presented by Molloy and Reed [21, Chapter 13]; the extended version of Bansal, Gupta, and Guruganesh [6, Sections 6–9] develops his recursive probability modifications and entropy estimates. Molloy [20, Theorems 1 and 2] obtained the triangle-free list bound with leading constant $1 + o(1)$ and gave a new proof of the clique-free bound, with explicit dependence on r .

Bernshteyn [8] extended Johansson’s triangle-free bound to correspondence coloring, and later extended Molloy’s asymptotic triangle-free bound and the clique-free bound to that setting [9, Theorems 1.3 and 1.5]. In particular, the previous general upper bound for fixed $r \geq 4$ was

$$\chi_{\text{DP}}(G) = O_r\left(\frac{\Delta \log \log \Delta}{\log \Delta}\right).$$

Davies, Kang, Pirot, and Sereni [11, Theorem 30] developed a framework based on local occupancy in the hard-core model and improved the leading constants. Przybyło [23, Theorem 10] also obtained bounds under clique exclusion in the graph whose vertices are list entries and whose edges are forbidden pairs, while retaining the maximum degree of the underlying graph as the degree parameter. These results retain the $\log \log \Delta$ factor for a general fixed forbidden clique.

Other families of forbidden subgraphs admit the sharper order. Alon, Krivelevich, and Sudakov [2] proved that a graph whose neighborhoods each span at most Δ^2/f edges, with $2 \leq f \leq \Delta^2$, has chromatic number $O(\Delta/\log f)$. Vu [27] extended this local-sparsity bound to list coloring. Alon, Krivelevich, and Sudakov also deduced $O_F(\Delta/\log \Delta)$ for every fixed almost bipartite forbidden graph F , meaning that deleting at most one vertex from F leaves a bipartite graph. Anderson, Bernshteyn, and Dhawan [3, 4] proved correspondence-coloring bounds with leading constants $1 + o(1)$ for bipartite forbidden graphs and $4 + o(1)$ for almost bipartite forbidden graphs. Their more general theorems assume that the graph of forbidden pairs excludes F and use its maximum degree in place of $\Delta(G)$.

More recently, Dhawan, Janzer, and Methuku [12, Theorems 1.4 and 1.5] proved $O_t(\Delta/\log \Delta)$ for ordinary coloring of $K_{t,t,t}$ -free graphs, and hence for every fixed three-colorable forbidden graph. They also obtained the independence bound $(1 - o(1))n \log d/d$ in terms of the average degree d . Their coloring theorem concerns ordinary coloring; it does not supply a list or correspondence bound. Theorem 1.1 removes the $\log \log \Delta$ factor for correspondence coloring with any fixed forbidden clique K_r , $r \geq 4$.

The structural part of our argument is related to the independence problem. Ajtai, Erdős, Komlós, and Szemerédi [1] asked for a lower bound of order $n \log d/d$ for K_r -free graphs of average degree d . Shearer [24, 25] obtained the asymptotic triangle-free bound and a clique-free bound with a $\log \log d$ loss. The companion manuscript *A logarithmic independence bound for clique-free graphs* [22] proves the logarithmic bound and develops the weighted multipliers, neighborhood walks, and vertex splitting used here. The recursive use of clique-free neighborhoods has a precedent in [1, Section 3]; entropy increments controlling distances between walk distributions have a precedent in [7, Section 2, Theorem 8]. We adapt these ingredients to a local adjustment statement that also bounds how many entries from each list an operation affects. We establish the required weighted

and dual estimates directly.

The remaining probabilistic ingredients also have established antecedents. We use the simultaneous coupling bound presented by Angel and Spinka [5], with its finite-state proof included. Concentration for sums in which each independent input affects only a bounded number of summands is the setting of Gavinsky, Lovett, Saks, and Srinivasan [17]. We derive the weighted form needed here from Finner’s generalized Hölder inequality [16] and Hoeffding’s exponential-moment bound [18]. The Lovász local lemma [14] then makes the local round estimates hold simultaneously.

1.2 The proof and its main ingredients

The proof constructs a partial coloring in small random rounds. At an uncolored vertex v , each available entry $i \in L(v)$ carries a nonnegative weight w_i . Its total weight $W_v = \sum_{i \in L(v)} w_i$ must remain bounded below, while the individual weights remain small. These two conditions guarantee many available entries. The round colors enough neighbors to make the remaining degree decrease faster than this supply of entries.

Let H be the graph whose vertices are the positive-weight entries and whose edges are the forbidden pairs in the matchings. The lists partition $V(H)$ into independent sets. Every vertex of H has at most one neighbor in each part, and every clique of H projects to a clique in G . Thus H is K_r -free. This graph allows us to work directly with the correspondences, without identifying entries from different lists.

A round independently activates each entry with probability proportional to its weight. An activated entry can be chosen if none of its neighbors in H activates. Choosing it removes all conflicting entries from the remaining lists. The potential controlling the loss of weight has three terms: an entropy relative to the initial weights, a linear penalty for lost total weight, and the incident edge mass

$$D_v = \sum_{i \in L(v)} \sum_{j \in N_H(i)} w_i w_j.$$

The last term measures the amount of conflicting weight around v . Deleting newly colored neighbors decreases it and pays for the cost of the round.

Triangles and local adjustments. An activation at a common neighbor of adjacent entries i, j removes both weights at once. Even after compensating their individual losses to first order, this shared deletion creates a positive edge correlation. Summed at an entry i , its cost is proportional to

$$t_i = w_i \sum_{j \in N_H(i)} w_j \sum_{z \in N_H(i) \cap N_H(j)} w_z.$$

This is twice the weighted mass of triangles through i . Lemma 2.1 supplies random local multiplications and deletions of weights whose combined contribution offsets t_i , up to a constant times the incident edge mass, simultaneously at all entries. The lemma also bounds the total rate of operations at each entry.

The structural argument develops the multiplier and entropy-splitting methods of the companion manuscript *A logarithmic independence bound for clique-free graphs* [22, Sections 2–5]. We prove all the ingredients needed here. Finite-dimensional separation tests the vertex inequalities and the rate bounds against arbitrary nonnegative prices. This reduces the adjustment problem to bounding the resulting priced triangle mass by a constant multiple of the priced edge mass, together with the allowance supplied by the rate bounds. Because vertex prices may vary widely, we group triangles by the dyadic size of their median price and rescale the weights within each

level. The dual constraints then bound the weight of edges between nearby sets. Walks inside neighborhoods guide a splitting of vertices into copies, preserving most triangle weight while sharply reducing every neighborhood weight. Repetition leaves bounded neighborhood weights, where the triangle-to-edge comparison follows directly. The accumulated losses are controlled independently of the number of price levels.

Controlling participation within each list. For correspondence coloring, a support of small graph radius may still contain many entries from one list. Radius alone therefore does not give the concentration needed for the coloring round. The adjustment lemma controls both quantities: each operation is supported in a ball, and it affects only a bounded number of entries in each part of the partition. When the neighborhood weight is $O(\log \Delta)$, both its multiplier bound and its participation bound are $\Delta^{o(1)}$.

The additional bound comes from the local sets used in the walk argument. A vertex is added to such a set only when it has a prescribed positive amount of neighbor weight in the preceding set. The one-neighbor-per-part property converts this admission threshold into a bound on the number of vertices added from each part. It survives splitting because the projection of the split graph is injective on directed edges. No lower bound on individual vertex weights is needed. This gives the adjustment lemma as a separate statement for partitioned weighted graphs, before its application to coloring.

Concentration and completion. The trial uses independent flags for activations and adjustments. Different entry weights after the trial need not be independent. Instead, Lemma 4.3 controls sums of functions of independent inputs when each input is used by few summands. The participation bound applies directly to the total mass and the non-edge part of the potential at a list. We control the edge part by concentrating each matching sum separately.

Activations may collide across forbidden pairs. To show that these collisions cost little, we first bound the degree of their activated subgraph. Too many collision edges would then contain a large matching, whose endpoint activations are independent. This supplies the required tail estimate without a decomposition into globally named colors. The Lovász local lemma gives simultaneous progress at all vertices. Trimming extreme weights restores the assumptions for the next round. Once every remaining degree is smaller than the number of available entries, a greedy coloring completes the assignment.

Section 2 states the adjustment lemma and develops the dual local estimates. Section 3 proves the entropy and splitting estimates and completes that lemma. Section 4 constructs the random update, proves its expectation and concentration bounds, and restores the weight conditions after simultaneous progress. Section 5 iterates this round and completes Theorem 1.1.

1.3 Conventions

Neighborhoods are open; graph-distance balls include their centers. An edge or triangle has weight equal to the product of its vertex weights. Unless indicated otherwise, sums over edges and triangles are unordered. A graph homomorphism is *injective on directed edges* if distinct ordered adjacent pairs have distinct images. It may identify vertices that do not occur in the same directed edge. Constants implicit in $O_r(\cdot)$ and sufficiently large thresholds may depend on r , but not on the graph, lists, or matchings. Auxiliary graphs may be empty.

2 Local weight adjustment with bounded participation

The coloring argument requires local changes of weights that offset the contribution of triangles. Each change must also involve few entries from any one list. We formulate this requirement for a partition of a weighted graph, with constants independent of the graph and of the size of its parts.

Fix an integer $r \geq 4$. Let H be a finite K_r -free graph with a partition $\mathcal{D}$ of its vertex set such that

$$|N_H(i) \cap D| \leq 1 \quad (i \in V(H), D \in \mathcal{D}).$$

Give each vertex i a positive weight w_i , and write

$$s_i = \sum_{u \in N_H(i)} w_u, \quad d_i = w_i s_i, \quad t_i = w_i \sum_{u \in N_H(i)} w_u \sum_{z \in N_H(u) \cap N_H(i)} w_z. \quad (2.1)$$

Thus d_i is the mass of edges incident to i , and t_i is twice the mass of triangles containing i . Suppose that $s_i \leq \Lambda$ for every i , where $\Lambda \geq 1$. Define

$$f(a) = a \log a - a + 1 \quad (a > 0), \quad f(0) = 1,$$

and put

$$\ell_0 = \log(8\Lambda), \quad R_0 = \lceil 4\ell_0 \rceil + 10, \quad U = \exp(A\ell_0^2), \quad P_0 = \lceil \exp(40\ell_0^2) \rceil, \quad (2.2)$$

where $A = A(r) > 0$ will be chosen below.

A *multiplier operation* consists of a nonempty listed support $J \subseteq V(H)$, contained in a ball of radius R_0 in H and satisfying $|J \cap D| \leq P_0$ for every $D \in \mathcal{D}$, together with multipliers $m_i \in [0, U]$ for $i \in J$. Extend $m_i = 1$ off J , and associate to the operation the vector

$$g_i = w_i f(m_i) + \sum_{u \in N_H(i)} w_i w_u (m_i - 1)(m_u - 1). \quad (2.3)$$

Multipliers equal to 1 are permitted on J . A *drop operation* at i has listed support $J = \{i\}$ and vector

$$g_i = \Lambda w_i - d_i, \quad g_u = -w_i w_u \quad (u \in N_H(i)), \quad g_u = 0 \quad (u \notin N_H(i) \cup \{i\}). \quad (2.4)$$

The listed support records the vertices on which the operation acts; a drop vector can have nonzero coordinates outside that support. It charges Λw_i for losing weight at i and subtracts its incident edge masses.

The two terms of (2.3) have a first-order interpretation. For fixed multipliers m_i and small $\rho > 0$, apply them on a common event of probability ρ , and use $(1 - \rho m_i)/(1 - \rho)$ otherwise. Each weight then has unchanged expectation. The expected change of the entropy term $w_i \log w_i - w_i$, with $0 \log 0 = 0$, is $\rho w_i f(m_i) + O(\rho^2)$, while the expected change on an edge iu is $\rho w_i w_u (m_i - 1)(m_u - 1) + O(\rho^2)$. In particular, events of probability ρw_z that delete the weights on $N_H(z)$, with this compensation of their mean loss, contribute ρd_i to the entropy change at i and ρt_i to its incident edge change, to first order. The following lemma supplies operations that offset the latter contribution simultaneously at all vertices.

Lemma 2.1 (Local weight adjustment). *For every integer $r \geq 4$ there are constants $A = A(r) > 0$, $B = B(r) \geq 1$, and $\Lambda_r \geq 1$ with the following property. Let $\Lambda \geq \Lambda_r$, and let H be a finite K_r -free graph whose vertex set has a partition $\mathcal{D}$ such that each vertex has at most one neighbor in each*

part. For any positive vertex weights with $s_i \leq \Lambda$, there is a finite family of the multiplier and drop operations defined above, with vectors $g^{(j)}$, listed supports J_j , and rates $a_j \geq 0$, such that

$$t_i + \sum_j a_j g_i^{(j)} \leq B d_i, \quad \sum_{j: i \in J_j} a_j \leq \Lambda^4 \quad (i \in V(H)). \quad (2.5)$$

The constants depend only on r .

Taking every part to be a singleton gives the same weight adjustment conclusion for an arbitrary positively weighted K_r -free graph, with no partition specified: the neighbor and support conditions on parts then hold automatically.

The proof occupies this section and Section 3. Nonnegative prices first reduce the simultaneous vertex inequalities to a scalar triangle estimate. We then derive a local edge estimate from the same prices. Section 3 uses that edge estimate to bound the triangles and finish the proof. If H has no vertices, the lemma holds with no operations, so we assume otherwise throughout the reduction.

2.1 Dual prices and dyadic levels

A bound on total triangle mass would not guarantee the individual vertex inequalities in (2.5). To retain those constraints, assign prices $x_i \geq 0$ to the first inequalities in (2.5), and prices $w_i y_i \geq 0$ to the incidence budgets in the second inequalities. It suffices to prove that

$$\sum_i x_i g_i + \sum_{i \in J} w_i y_i \geq 0 \quad \text{for every allowed operation } (g, J) \quad (2.6)$$

implies

$$\sum_i x_i t_i \leq B \sum_i x_i d_i + \Lambda^4 \sum_i w_i y_i. \quad (2.7)$$

The support price in (2.6) includes every vertex of the listed support, even when its multiplier equals 1.

To justify the reduction, let $n = |V(H)|$ and let $\mathcal{C}$ be the cone in $\mathbb{R}^{2n}$ generated by the operation columns $(g, \mathbf{1}_J)$ and the nonnegative orthant. These columns form a compact set: there are finitely many possible supports, each multiplier cube is compact, and f is continuous on $[0, U]$. Moreover, the sum of the last n coordinates of every column is at least 1. In any convergent sequence of elements of $\mathcal{C}$, the last-coordinate sums therefore bound the total coefficients of the operation columns. Their combined contributions belong to a fixed multiple of the compact convex hull of the columns and zero. A subsequence of those contributions converges; the residual vectors then converge in the nonnegative orthant. This proves that $\mathcal{C}$ is closed.

Membership of $(Bd - t, \Lambda^4 \mathbf{1})$ in $\mathcal{C}$ is exactly the existence of the finite family in (2.5). If this target were outside the cone, separation would give a linear functional nonnegative on $\mathcal{C}$ and negative at the target. Since $\mathcal{C}$ contains the nonnegative orthant, the coefficients of that functional are nonnegative. Because $w_i > 0$, they may be written as $(x_i, w_i y_i)_i$. They would satisfy (2.6) and violate (2.7), proving the reduction.

Fix nonnegative prices satisfying (2.6). Applying that inequality to the drop at i and dividing by w_i gives

$$\sum_{u \in N_H(i)} x_u w_u \leq (\Lambda - s_i) x_i + y_i \leq \Lambda x_i + y_i. \quad (2.8)$$

Call a vertex i *good* if $x_i > y_i$, and *bad* otherwise. A triangle $\{i, u, z\}$ contributes $2(x_i + x_u + x_z)w_iw_uw_z$ to $\sum_v x_v t_v$. For a fixed bad vertex i , the sum of these contributions over triangles containing i is at most

$$2w_i \left(x_i s_i^2 + s_i \sum_{u \in N_H(i)} x_u w_u \right) \leq O(\Lambda^2) w_i y_i,$$

by (2.8) and $s_i \leq \Lambda$. Summing over bad vertices, with possible overcounting, yields

$$\sum_{\substack{\{i,u,z\} \text{ triangle of } H \\ \{i,u,z\} \text{ contains a bad vertex}}} 2(x_i + x_u + x_z)w_iw_uw_z \leq O(\Lambda^2) \sum_i w_i y_i. \quad (2.9)$$

The remaining triangles have only good vertices. Their edge budget is

$$M_x = \sum_i x_i d_i = \sum_{iu \in E(H)} (x_i + x_u) w_i w_u, \quad (2.10)$$

where the edge sum is unordered.

Good vertices have positive prices. Consider all dyadic numbers $b = 2^j$, $j \in \mathbb{Z}$, such that $[b, 2b)$ contains the price of a good vertex. There are only finitely many such levels. At level b , call a good vertex low if $x_i < b$, middle if $b \leq x_i < 2b$, and high if $x_i \geq 2b$. Let H_b be the graph on good vertices obtained from H by deleting edges with two low endpoints or two high endpoints. Rescale the weights to

$$q_i = \begin{cases} w_i x_i / b, & i \text{ high,} \\ w_i, & i \text{ low or middle.} \end{cases} \quad (2.11)$$

The weights q and all three classifications refer to the level under consideration.

A triangle of good vertices survives at exactly one level: the level whose interval contains its median price. At that level it has at most one low and at most one high vertex, and hence at least one middle vertex. At every other level, two vertices are low or two are high, and the edge between them is deleted. The rescaling pays for the largest price even when it is much larger than the median: if $x_i < b \leq x_u < 2b \leq x_z$, then

$$b q_i q_u q_z = x_z w_i w_u w_z.$$

If a retained triangle has no high vertex, all three prices are less than $2b$. If it has a high vertex, its q -weight incorporates that largest price divided by b . In either case its contribution satisfies

$$2(x_i + x_u + x_z)w_iw_uw_z \leq 12b q_i q_u q_z.$$

Define

$$T_0 = \sum_b b \sum_{\{i,u,z\} \text{ triangle of } H_b} q_i q_u q_z. \quad (2.12)$$

We have proved

$$\sum_i x_i t_i \leq 12T_0 + O(\Lambda^2) \sum_i w_i y_i. \quad (2.13)$$

Consequently, a bound $T_0 = O_r(M_x)$ will imply (2.7) once B and the lower threshold on Λ are chosen.

To obtain that bound, we express the dual prices in the rescaled weights. For good vertices and original edges between them, set

$$\alpha_i = \frac{x_i w_i}{b q_i}, \quad \beta_i = \frac{y_i w_i}{b q_i}, \quad \kappa_{iu} = \frac{(x_i + x_u) w_i w_u}{b q_i q_u}. \quad (2.14)$$

They satisfy

$$0 \leq \beta_i < \alpha_i \leq 2, \quad 0 < \kappa_{iu} \leq 4, \quad \kappa_{iu} \geq 1 \quad (iu \in E(H_b)). \quad (2.15)$$

Indeed, $\alpha_i = 1$ at high vertices and $\alpha_i = x_i/b$ at the others. If exactly one endpoint of an edge is high, then κ_{iu} is 1 plus the ratio of the other price to the high price. If both endpoints are high it is $b/x_i + b/x_u$, and if neither is high it is $(x_i + x_u)/b$. A retained edge of the last type has a middle endpoint, proving its lower bound.

For a multiplier operation supported on good vertices, division of (2.6) by b gives

$$\begin{aligned} 0 \leq & \sum_{i \text{ good}} \alpha_i q_i f(m_i) + \sum_{\substack{iu \in E(H) \\ i, u \text{ good}}} \kappa_{iu} q_i q_u (m_i - 1)(m_u - 1) \\ & + \sum_{i \in J} \beta_i q_i. \end{aligned} \quad (2.16)$$

The edge sum here still uses the original edges of H , including edges absent from H_b .

The rescaling also preserves a uniform neighborhood bound:

$$\sum_{u \in N_{H_b}(i)} q_u \leq e^{\ell_0} = 8\Lambda. \quad (2.17)$$

For a high vertex, all retained neighbors have their original weights, so the sum is at most Λ . For a non-high vertex i , the high neighbors have total q -weight at most

$$\frac{1}{b} \sum_{u \in N_H(i)} x_u w_u \leq \frac{\Lambda x_i + y_i}{b} < 2(\Lambda + 1),$$

by (2.8) and goodness. The other neighbors contribute at most Λ , proving (2.17).

2.2 Sparse multipliers and local edge mass

The normalized dual inequality controls the mass of edges between two nearby sets. To see this, we will increase the weights on one set and set the weights on the other to zero. The cross edges then have negative quadratic contribution. The following construction keeps the positive contribution inside the first set small. Its recursive passage to neighborhoods excluding a smaller clique is the sparse-subgraph device of Ajtai, Erdős, Komlós, and Szemerédi [1, Section 3]. The mean-one weighted construction and the resulting cross estimate adapt [22, Lemmas 2.2–2.3]; we give the proofs with the locality and participation conditions needed here.

Lemma 2.2 (Sparse multipliers). *For every integer $r \geq 2$ there is a constant $D_r \geq 1$ such that the following holds. Let F be a finite K_r -free graph with positive vertex weights q_i of total mass s . For every $0 < \epsilon \leq 1/2$, there are jointly distributed random multipliers n_i satisfying*

$$0 \leq n_i \leq D_r \epsilon^{-(r-2)}, \quad \mathbb{E} n_i = 1, \quad \mathbb{E} \sum_{iu \in E(F)} q_i q_u n_i n_u \leq \epsilon s^2, \quad (2.18)$$

where the edge sum is unordered.

Proof. The assertion is vacuous if F has no vertices. Otherwise $s > 0$, and we argue by induction on r . For $r = 2$, the graph has no edges and $n_i = 1$ suffices.

Keep the initial total mass s fixed. Successively remove the open neighborhood of a vertex in the current remainder whenever that neighborhood has mass greater than $\epsilon s/4$. The removed sets are disjoint, each induces a K_{r-1} -free graph, and their masses s_j satisfy $s_j > \epsilon s/4$. In the final remainder, every neighborhood has mass at most $\epsilon s/4$, so its unordered edge mass is at most $\epsilon s^2/8$. If no set was removed, the constant multipliers 1 already prove the assertion.

Otherwise, select the final remainder with probability $1/2$, or removed set j with probability

$$p_j = \frac{s_j}{2 \sum_l s_l} > \frac{\epsilon}{8}.$$

Set the multipliers to zero off the selected set. On the selected remainder, use the constant multiplier 2. On a selected removed set j , use the inductively supplied multipliers with parameter $\epsilon/4$, divided by p_j . Every vertex has mean multiplier 1. On a removed set, the multiplier bound follows from

$$\frac{D_{r-1}(\epsilon/4)^{-(r-3)}}{p_j} \leq 8 \cdot 4^{r-3} D_{r-1} \epsilon^{-(r-2)}.$$

Thus a constant D_r depending only on r bounds all multipliers as required.

Edges between different sets contribute zero. The remainder contributes at most $\epsilon s^2/4$ in expectation, while the removed sets together contribute at most

$$\sum_j \frac{(\epsilon/4)s_j^2}{p_j} = \frac{\epsilon}{2} \left(\sum_j s_j \right)^2 \leq \frac{\epsilon}{2} s^2.$$

Their sum is at most ϵs^2 , completing the induction. $\square$

For sets S, D in a graph with weights q , write

$$q(S) = \sum_{i \in S} q_i, \quad e_q(S, D) = \sum_{i \in S} \sum_{u \in D \cap N(i)} q_i q_u.$$

This is a directed edge sum: an edge with both endpoints in $S \cap D$ is counted twice. We specify the graph by a subscript when needed.

Lemma 2.3 (Local cross estimate). *Assume the prices satisfy (2.6), and fix a level b . Let $S, D \subseteq V(H_b)$, where $S \cup D$ is contained in a ball of radius R_0 in H and meets every part of $\mathcal{D}$ in at most P_0 vertices. Suppose that*

$$32 \leq h \leq 20\ell_0^2, \quad q(S), q(D) \leq e^h, \quad 0 \leq \eta \leq 2,$$

and that $\alpha_i, \beta_i \leq \eta$ for $i \in S$. If A in (2.2) is sufficiently large depending only on r , then

$$e_{q, H_b}(S, D) \leq C_1(\eta h q(S) + e^{-3h}), \quad (2.19)$$

where $C_1 = C_1(r)$.

Proof. First suppose that S, D are disjoint and nonempty. Apply Lemma 2.2 to the original induced graph $H[S]$, with its q -weights and $\epsilon = e^{-16h}$. Set $k' = e^{8h}$, and take the multiplier operation with listed support $S \cup D$ and

$$m_i = 1 + k' n_i \quad (i \in S), \quad m_i = 0 \quad (i \in D).$$

The sparse multiplier bound implies $\log(1 + k'n_i) \leq C'_r h$. Since $h \leq 20\ell_0^2$, choosing $A \geq 20C'_r$ makes every realization an allowed operation; the ball and participation conditions follow from the hypotheses on $S \cup D$.

Write $s = q(S)$ and $d = q(D)$, and take expectations in (2.16). On S , the expected entropy term is at most $C'_r \eta h k' s$, because $f(1 + a) \leq a \log(1 + a)$ for $a \geq 0$ and $\mathbb{E}n_i = 1$. The support price on S is at most ηs . On D , the entropy and support terms together are at most $4d$, using $f(0) = 1$ and (2.15).

The expected quadratic edge term is at most

$$4(k')^2 \epsilon s^2 + 2d^2 - k' e_{q, H_b}(S, D). \quad (2.20)$$

Inside S , the bound $\kappa_{iu} \leq 4$ and (2.18) give the first term, including for original edges absent from H_b . Inside D , the unordered edge mass is at most $d^2/2$, giving the second term. On retained edges between S and D , the coefficient is at least 1 and the expected product of deviations is $-k'$. Deleted cross edges have nonpositive contributions and can be omitted in an upper bound. Edges leaving $S \cup D$ contribute zero.

Since the tested inequality holds for every realization, these bounds give

$$e_{q, H_b}(S, D) \leq C'_r \eta h s + \frac{\eta s + 4d + 4(k')^2 \epsilon s^2 + 2d^2}{k'} \leq C'_r \eta h s + O(e^{-3h}).$$

For the last inequality, $s, d \leq e^h$, $\eta \leq 2$, $k' = e^{8h}$, and $(k')^2 \epsilon = 1$ make the numerator after division at most $6e^{-7h} + 6e^{-6h}$. If either set is empty, the bound holds immediately.

For possibly overlapping sets, assign every vertex independently to a left or right class with equal probabilities. Apply the disjoint estimate to the part of S on the left and the part of D on the right. Each directed edge of $e_{q, H_b}(S, D)$ survives with probability $1/4$, whereas the expected mass of the left subset of S is $q(S)/2$. Taking expectations and increasing C_1 proves (2.19) in general. $\square$

The triangle argument will replace vertices by copies and delete some edges. The next observation ensures that the local estimate remains available after these changes. A homomorphism $p : F \rightarrow H_b$ is *injective on directed edges* if the map $(u, v) \mapsto (p(u), p(v))$ is injective on all ordered adjacent pairs of F . Vertices may have several preimages, but each original directed edge has at most one preimage.

Corollary 2.4 (Transfer under vertex splitting). *Let F be a graph with a homomorphism $p : F \rightarrow H_b$ that is injective on directed edges. Give every vertex of F the weight, prices, coefficients, and part of its image. Then every vertex of F has at most one neighbor in each inherited part. Moreover, (2.19) holds in F for any $S, D \subseteq V(F)$ satisfying the mass and coefficient hypotheses of Lemma 2.3, provided $S \cup D$ lies in a ball of radius R_0 in F and meets each inherited part in at most P_0 vertices.*

Proof. Directed-edge injectivity implies

$$e_{q, F}(S, D) \leq e_{q, H_b}(p(S), p(D)), \quad q(p(S)) \leq q(S), \quad q(p(D)) \leq q(D).$$

The projected union lies in a ball of radius R_0 in H and meets each original part in at most P_0 vertices. The coefficient bounds also pass to the images. Lemma 2.3, which permits overlap of the two sets, therefore proves the asserted estimate. Finally, distinct neighbors of a vertex of F have distinct images: otherwise two directed edges would have the same image. Since their images lie in the neighborhood of one vertex of H_b , at most one belongs to any given part. $\square$

We have derived the local edge estimate solely from the dual condition and have shown that it survives the graph modifications needed below. The remaining assertion is the following triangle bound.

Proposition 2.5 (Triangle bound). *For the graphs H_b , weights q , and prices constructed above from (2.6), with A chosen as in Lemma 2.3, the quantities in (2.10) and (2.12) satisfy*

$$T_0 \leq O_r(M_x).$$

The implicit constant depends only on r .

Section 3 proves the proposition by repeatedly reducing neighborhood weights while retaining a controlled fraction of the triangle mass. In view of (2.13), this completes the dual inequality and Lemma 2.1. The constants are chosen in this order: the sparse multiplier construction determines A and C_1 ; the triangle bound then determines B ; and finally Λ_r is increased so that the $O(\Lambda^2) \sum_i w_i y_i$ term is at most $\Lambda^4 \sum_i w_i y_i$.

3 Controlling triangles by splitting vertices

We prove Proposition 2.5 by repeatedly reducing the neighborhood weights in the level graphs. The main construction splits a vertex into copies of the same weight, distributes its incident edges among those copies, and preserves most of the triangle mass. Two features must hold together: the edges of a typical triangle must meet at the same copy of each corner, and each copy must have a much smaller neighborhood weight. Random walks inside neighborhoods supply labels that achieve both properties. The argument adapts the triangle estimate of [22, Sections 3–5], while retaining the bounds on radius and participation in each part needed for the weight adjustments here.

Throughout this section, a graph at level b has a homomorphism into H_b that is injective on directed edges. Its vertices inherit the weights q , prices, parts, and low, middle, or high classification of their images. Corollary 2.4 therefore makes (2.19) available on each such graph, with its stated ball and part conditions. Deleting vertices or edges preserves these properties. The splitting construction will preserve them as well.

3.1 Removing triangles of small total mass

Let F_b denote the current graph at level b , initially H_b , and put

$$T = \sum_b \sum_{\{a,u,v\} \text{ a triangle of } F_b} q_a q_u q_v.$$

Thus initially $T = T_0$. A triangle of F_b projects to a triangle of H_b with three distinct vertices and hence has a middle vertex. We shall use the following bound at every stage:

$$\sum_b \sum_{\substack{a \in V(F_b) \\ a \text{ middle}}} q_a q(N_{F_b}(a)) \leq 2M_x. \quad (3.1)$$

Indeed, for an edge au the quantity $bq_a q_u$ is at most the contribution of its image edge to M_x , since the coefficient κ on that image is at least one. An original vertex is middle at exactly one level, and a directed edge at that level has at most one preimage. Each original edge can consequently be charged at most once from each endpoint.

Suppose that all current neighborhood weights are at most e^ℓ , where $32 \leq \ell \leq \ell_0$. First delete, at every level, the vertices whose inherited prices satisfy $x_i/b < \ell^{-4}$. To bound the triangle mass lost, fix a middle vertex a before these deletions, let $D = N_{F_b}(a)$, and let S be the set of vertices of D being deleted. The set $S \cup D = D$ lies in a ball of radius one and has at most one vertex in each part. Also $q(S), q(D) \leq e^\ell$, while $\alpha_i, \beta_i \leq \ell^{-4}$ on S . Applying (2.19) with $h = \ell$ gives

$$e_q(S, D) \leq C_1(\ell^{-3}q(D) + e^{-3\ell}).$$

In fact $e_q(S, D) = O_r(q(D)/\ell^2)$: when $q(D) > \ell^{-2}$ the exponential term is absorbed by $q(D)/\ell^2$; when $q(D) \leq \ell^{-2}$, use instead $e_q(S, D) \leq q(D)^2$. A deleted triangle has a middle vertex, which is not itself deleted, and an edge from S to D in that middle vertex's neighborhood. Multiplying by bq_a and summing therefore bounds the loss in T by $O_r(M_x/\ell^2)$, using (3.1).

After these deletions, the total edge mass over the levels satisfies

$$\sum_b b \sum_{uv \in E(F_b)} q_u q_v \leq O(\log \ell) M_x. \quad (3.2)$$

To see this, write $x_{\min}$ for the smaller of the two prices on an original edge. If that edge has a preimage at level b , then

$$b\ell^{-4} \leq x_{\min} < 2b.$$

The first inequality follows from the deletion rule, and the second from the absence of edges with two high endpoints. There are only $O(\log \ell)$ dyadic choices for b . At each such level, the edge has at most one preimage, whose mass $bq_u q_v$ is at most the original edge's contribution to M_x .

Next delete an edge whenever the weight of its current common neighborhood is less than ℓ^{-1} , continuing until no such edge remains. Deleting uv destroys triangle mass at most $bq_u q_v / \ell$. Thus (3.2) bounds this further loss by $O((\log \ell) M_x / \ell)$. Discard isolated vertices as well.

Fix a nonempty level graph $F = F_b$ after these operations. All neighborhoods in the next two subsections are in F . Set

$$\begin{aligned} S_u &= q(N(u)) \in [\ell^{-1}, e^\ell], \\ c_{uv} &= q(N(u) \cap N(v)) \geq \ell^{-1} \quad (uv \in E(F)), \\ h_u &= \sum_{v \in N(u)} q_v c_{uv} \geq S_u / \ell. \end{aligned} \quad (3.3)$$

The lower bound on S_u follows because u has a neighbor and every edge has common-neighborhood weight at least ℓ^{-1} .

Lemma 3.1 (Splitting a level graph). *Let $32 \leq \ell \leq \ell_0$, and let F be a nonempty finite graph with no isolated vertices, with a homomorphism into some H_b injective on directed edges. Give its vertices the inherited weights, prices, and parts, and suppose that (3.3) holds. There is a graph F' obtained by splitting vertices and retaining at most one copy of each edge of F , with unchanged vertex weights, such that every neighborhood of F' has weight at most $e^{\sqrt{\ell}}$ and*

$$\sum_{\{a,u,v\} \text{ a triangle of } F'} q_a q_u q_v \geq \left(1 - C_2 \frac{\log \ell}{\sqrt{\ell}}\right) \sum_{\{a,u,v\} \text{ a triangle of } F} q_a q_u q_v, \quad (3.4)$$

where C_2 depends only on r . The projection $F' \rightarrow F$ is a homomorphism injective on directed edges.

We prove the lemma in the next two subsections. At a vertex u , the construction will attach a label in $N(u)$ to each incidence (u, v) ; edges with the same label use the same copy of u . A triangle survives only if its two labels agree at each corner. We first construct probability laws whose labels can usually agree, then impose a condition on the reverse transition probabilities to bound the weight at each copy.

3.2 Entropy bounds for walks in neighborhoods

Let t be the unordered triangle mass of F . It is positive by (3.3). Choose an ordered triangle (u, v, z) with probability $q_u q_v q_z / (6t)$. The law of its first two vertices and the conditional law of its second vertex are

$$\mu(u, v) = \frac{q_u q_v c_{uv}}{6t}, \quad \pi_u(v) = \frac{q_v c_{uv}}{h_u}. \quad (3.5)$$

In particular, the marginal law of u is $q_u h_u / (6t)$, and μ is symmetric in its two arguments.

On $N(u)$ define the transition kernels

$$P_u(v, z) = \frac{\mathbf{1}_{\{vz \in E(F)\}} q_z}{c_{uv}}, \quad Q_u = \frac{I + P_u}{2}, \quad K_u^{(j)} = P_u Q_u^j \quad (j \geq 0). \quad (3.6)$$

All denominators are positive. The kernel P_u is reversible with stationary law π_u , since

$$\pi_u(v) P_u(v, z) = \frac{q_v q_z \mathbf{1}_{\{vz \in E(F)\}}}{h_u}$$

is symmetric in v, z . The same holds for Q_u and $K_u^{(j)}$, which are polynomials in P_u . Moreover, writing $K_u^{(j)} = Q_u^j P_u$ and using $c_{uv} \geq \ell^{-1}$ gives

$$K_u^{(j)}(v, z) \leq \ell q_z. \quad (3.7)$$

The number of vertices in $N(u)$ can be arbitrarily large even when its total weight is small. We therefore measure entropy relative to q : for a probability row p on a finite set with positive weights q , put

$$\mathcal{H}_q(p) = \sum_{z: p(z) > 0} p(z) \log \frac{q_z}{p(z)}.$$

This quantity may be negative. Jensen's inequality bounds it by $\log q(\text{supp } p)$. Define

$$H_j = \mathbb{E}_{(u,v) \sim \mu} \mathcal{H}_q(K_u^{(j)}(v, \cdot)), \quad R = \lceil \ell \rceil. \quad (3.8)$$

The density bound gives $H_j \geq -\log \ell$. We will bound H_R by $O_r(\log \ell)$, much less than the immediate bound $H_R \leq \ell$. Only $O_r(\log \ell)$ entropy can then be gained over R steps, so some step has a small increment. That increment will control the distance between the label laws at a triangle corner.

Lemma 3.2. *With the above definitions,*

$$H_0 \geq -\log \ell, \quad H_R \leq O_r(\log \ell). \quad (3.9)$$

Proof. Only the upper bound remains. Fix v . We construct sets that contain almost all the mass of the walks starting at v , uniformly over the choice of $u \in N(v)$. Let

$$\theta = \ell^{-6} e^{-\ell}, \quad D_0 = N(v), \quad D_{j+1} = D_j \cup \{z : q(N(z) \cap D_j) \geq \theta\} \quad (0 \leq j < R).$$

The sets D_j lie in the ball of radius $j + 1$ about v . Double counting the edges between D_j and its newly included vertices yields

$$\theta q(D_{j+1} \setminus D_j) \leq e^\ell q(D_j).$$

Consequently,

$$q(D_R) \leq e^\ell (1 + \ell^6 e^{2\ell})^R \leq e^{20\ell^2}. \quad (3.10)$$

For the last inequality, use $6 \log \ell \leq \ell$ and $R \leq 2\ell$ for $\ell \geq 32$.

We also need the part condition in (2.19). For a fixed part D' , every vertex in $(D_{j+1} \setminus D_j) \cap D'$ has at least θ weight of neighbors in D_j . Each vertex of D_j has at most one neighbor in D' , so

$$\theta |(D_{j+1} \setminus D_j) \cap D'| \leq q(D_j).$$

Thus the weighted inclusion threshold controls an unweighted count in each part. The initial set $N(v)$ has at most one vertex in D' . It follows from (3.10) that

$$|D_R \cap D'| \leq 1 + R\theta^{-1} e^{20\ell^2} \leq e^{40\ell^2} \leq P_0. \quad (3.11)$$

The radius condition also holds, since $R + 1 \leq R_0$.

For every $u \in N(v)$, the leakage outside these sets satisfies

$$K_u^{(j)}(v, N(u) \setminus D_j) \leq j\ell^{-4} \quad (0 \leq j \leq R). \quad (3.12)$$

At $j = 0$, the row is supported on $N(u) \cap N(v)$. For the induction step, fix $z \in N(u) \setminus D_{j+1}$. Since $D_j \subseteq D_{j+1}$, mass arriving at z from D_j uses an off-diagonal transition of Q_u . By (3.7) and the off-diagonal bound $Q_u(y, z) \leq \ell q_z$, this mass is at most

$$\sum_{y \in N(z) \cap D_j} \ell q_y \ell q_z \leq \ell^2 q_z \theta = q_z \ell^{-4} e^{-\ell}.$$

Summing over these z gives at most ℓ^{-4} , since $q(N(u)) \leq e^\ell$. The mass arriving from outside D_j is at most the preceding leakage, proving (3.12).

Although D_R can have large total weight, its intersection with most neighborhoods relevant to the triangle law is small. Define

$$J_v = \{u \in N(v) : q(N(u) \cap D_R) > \ell^8\}.$$

Apply (2.19) to $N(v)$ and D_R , with $h = 20\ell^2$ and $\eta = 2$. The mass, radius, and part conditions were just verified; also $h \leq 20\ell_0^2$. As $S_v \geq \ell^{-1}$, the exponential error can be absorbed, giving

$$\ell^8 q(J_v) \leq e_q(N(v), D_R) \leq O_r(\ell^2 S_v), \quad q(J_v) \leq O_r(\ell^{-6} S_v).$$

This controls q -mass, whereas the conditional law given v in (3.5) also weights u by c_{uv} . A second use of (2.19), now on $J_v, N(v)$ with $h = \ell$ and $\eta = 2$, makes precisely this conversion:

$$\begin{aligned} \mathbb{P}_\mu(u \in J_v \mid v) &= \frac{\sum_{u \in J_v} q_u c_{uv}}{h_v} = \frac{e_q(J_v, N(v))}{h_v} \\ &\leq O_r\left(\frac{\ell q(J_v) + e^{-3\ell}}{S_v/\ell}\right) \leq O_r(\ell^{-4}). \end{aligned} \quad (3.13)$$

Here the sets are contained in $N(v)$, so the support conditions hold with radius one and at most one vertex per part.

For $u \notin J_v$, split the row $K_u^{(R)}(v, \cdot)$ according to membership in D_R . Its outside probability is at most $R\ell^{-4}$ by (3.12); the weights of the two possible supports are at most ℓ^8 and e^ℓ , respectively. The entropy of this mixture is therefore at most

$$\log 2 + 8 \log \ell + (R\ell^{-4})\ell.$$

Indeed, conditioning on the two supports adds their binary entropy, which is at most $\log 2$, and the conditional entropies are bounded by the logarithms of their support weights. Empty supports contribute nothing. For $u \in J_v$, use the bound ℓ . Averaging first over u conditional on v , using (3.13), and then over v proves $H_R = O_r(\log \ell)$. $\square$

3.3 Coupled labels and the split graph

Proof of Lemma 3.1. We first turn the entropy bound into similarity of the label laws at a triangle corner. For two probability rows p, a on the same finite set, write

$$\text{KL}(p||a) = \sum_{z:p(z)>0} p(z) \log \frac{p(z)}{a(z)}, \quad \text{TV}(p, a) = \frac{1}{2} \sum_z |p(z) - a(z)|,$$

with $\text{KL}(p||a) = +\infty$ when necessary. We shall use

$$\text{TV}(p, a) \leq \sqrt{\text{KL}(p||a)}. \quad (3.14)$$

For completeness, let $\rho = \sum_z \sqrt{p(z)a(z)}$. Jensen's inequality gives $\text{KL}(p||a) \geq -2 \log \rho \geq 2(1 - \rho)$, while Cauchy-Schwarz gives $\text{TV}(p, a)^2 \leq 1 - \rho^2 \leq 2(1 - \rho)$. The case of infinite divergence is immediate.

Since $K_u^{(j+1)} = Q_u K_u^{(j)}$, its row at v is the mixture of the rows $K_u^{(j)}(z, \cdot)$ with $z \sim Q_u(v, \cdot)$. The entropy of a mixture minus the average entropy of its constituent rows is the average of their divergences from the mixture. The $\log q_z$ terms cancel in this identity. Averaging over μ , and using stationarity of π_u for Q_u to identify the average constituent entropy as H_j , gives

$$H_{j+1} - H_j = \mathbb{E}_{\substack{(u,v) \sim \mu \\ z \sim Q_u(v, \cdot)}} \text{KL}(K_u^{(j)}(z, \cdot) || K_u^{(j+1)}(v, \cdot)) \geq 0. \quad (3.15)$$

Lemma 3.2 and telescoping over R steps now give an index $0 \leq j < R$ such that

$$H_{j+1} - H_j \leq O_r((\log \ell)/\ell).$$

Fix this index and abbreviate $K_u = K_u^{(j)}$. Using the row $K_u^{(j+1)}(v, \cdot)$ as an intermediary in the triangle inequality for total variation yields

$$\mathbb{E}_{\substack{(u,v) \sim \mu \\ z \sim P_u(v, \cdot)}} \text{TV}(K_u(v, \cdot), K_u(z, \cdot)) \leq 2\sqrt{H_{j+1} - H_j} \leq O_r\left(\sqrt{\frac{\log \ell}{\ell}}\right). \quad (3.16)$$

To check the factor two, the two expected distances to the intermediary sum to twice the expected distance from the z -row under $Q_u = (I + P_u)/2$. Apply (3.14), then Cauchy-Schwarz, and finally (3.15). The law on the left of (3.16) is exactly the ordered triangle law, because

$$\mu(u, v)P_u(v, z) = \frac{q_u q_v q_z}{6t} \quad \text{when } u, v, z \text{ form a triangle.}$$

At each u , give all incidences (u, v) labels $Y_{u,v} \in N(u)$, with marginal law $K_u(v, \cdot)$, by a simultaneous coupling. We use the disagreement bound of Angel and Spinka [5, Theorem 2]; the following finite-set construction includes the proof needed here. For any finite family of laws on a finite set, take one shared sequence of independent proposals, each consisting of a uniformly chosen label and an independent uniform mark in $[0, 1]$. Each law accepts the first proposal whose mark is no greater than its probability of that label. The accepted label has the prescribed law, and every law accepts almost surely. For two laws at total variation distance d , the first proposal accepted by either is accepted by both with probability $(1 - d)/(1 + d)$: the sums of the pointwise minimum and maximum of their probabilities are $1 - d$ and $1 + d$. Their labels thus disagree with probability at most $2d/(1 + d) \leq 2d$. A single proposal sequence gives this bound simultaneously for every pair of laws. Applying the construction to the rows of K_u and using (3.16) bounds the average probability of a disagreement at a triangle corner by $O_r(\sqrt{(\log \ell)/\ell})$. Couplings at distinct vertices may be chosen independently.

Agreement controls triangle retention. To control neighborhood weights, call an incidence (u, v) with label $y = Y_{u,v}$ *qualified* if

$$K_u(y, v) \geq q_v e^{-\sqrt{\ell}}. \quad (3.17)$$

This uses the reverse transition: at a copy indexed by y , summing $K_u(y, v)$ over its possible neighbors will sum a single probability row. It remains to show that this qualification rarely discards an incidence.

Choose $(u, v) \sim \mu$ and then $y \sim K_u(v, \cdot)$. Conditional on u , the joint law of (v, y) is $\pi_u(v)K_u(v, y)$, which is symmetric by reversibility. In particular, the reverse entry is positive almost surely, and exchanging v and y gives

$$\mathbb{E} \log \frac{q_v}{K_u(y, v)} = \mathbb{E} \log \frac{q_y}{K_u(v, y)} = H_j \leq H_R. \quad (3.18)$$

The random logarithm is at least $-\log \ell$ by (3.7). Add $\log \ell$ and apply Markov's inequality to this nonnegative variable. Lemma 3.2 then yields

$$\mathbb{P}((u, v) \text{ is not qualified}) \leq \frac{H_j + \log \ell}{\sqrt{\ell} + \log \ell} \leq O_r\left(\frac{\log \ell}{\sqrt{\ell}}\right). \quad (3.19)$$

For each u make copies (u, y) indexed by $y \in N(u)$, all with weight q_u . Retain uv if its two incidences qualify, and join $(u, Y_{u,v})$ to $(v, Y_{v,u})$. This defines a simple graph, since each old edge contributes at most one edge and its endpoints project to distinct vertices. The neighbors of (u, y) have total weight at most

$$\sum_{\substack{v \in N(u) \\ Y_{u,v}=y, uv \text{ retained}}} q_v \leq e^{\sqrt{\ell}} \sum_{v \in N(u)} K_u(y, v) = e^{\sqrt{\ell}}, \quad (3.20)$$

by (3.17). Figure 1 illustrates how agreement or disagreement of the corner labels determines whether a triangle survives.

An old triangle survives with its original mass if all six incidences qualify and the two labels agree at each of its three corners. In the ordered triangle law, each incidence has marginal μ , and each corner has the law used in (3.16). A union bound, followed by averaging with the triangle weights, shows that the expected fraction of triangle mass lost is at most

$$O_r\left(\frac{\log \ell}{\sqrt{\ell}} + \sqrt{\frac{\log \ell}{\ell}}\right) = O_r\left(\frac{\log \ell}{\sqrt{\ell}}\right).$$

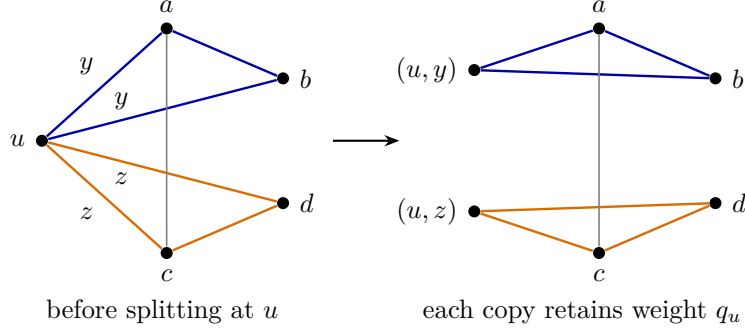


Figure 1: A local view of the splitting construction. All displayed incidences are assumed to qualify, and labels at vertices other than u are suppressed. These are auxiliary incidence labels, not colors in the correspondence assignment. The incidences ua, ub have label y , while uc, ud have label $z \neq y$. The triangles uab and ucd remain triangles after splitting at u ; uac does not. Each old edge is used once, and each copy of u keeps its original weight. The coupling makes unequal labels at a triangle corner rare in weighted average.

Choose a realization retaining at least the expected lower bound. The projection to F is a homomorphism injective on directed edges, and composing it with the old map to H_b preserves that property. All vertex weights and inherited data remain unchanged. This proves (3.4) and the lemma. $\square$

3.4 Iteration and completion of the adjustment lemma

Proof of Proposition 2.5. Apply the deletions of Subsection 3.1 to all levels, then Lemma 3.1 to each nonempty level graph. Empty level graphs need no construction. The resulting graphs have neighborhood weights at most $e^{\sqrt{\ell}}$ and still have homomorphisms into their original H_b injective on directed edges. Combining the additive losses from deletion with (3.4) gives

$$T_{\text{new}} \geq \left(1 - C_2 \frac{\log \ell}{\sqrt{\ell}}\right) T - C_3 \frac{\log \ell}{\ell} M_x, \quad (3.21)$$

where the constants depend only on r . If the coefficient of T is negative, the inequality follows simply from $T_{\text{new}} \geq 0$.

Choose a sufficiently large constant $X \geq 32$, depending only on r . Starting at ℓ_0 , perform the reduction while $\ell > X$, replacing ℓ by $\sqrt{\ell}$ each time. Both losses in (3.21) are summable uniformly in the initial value ℓ_0 . To verify this explicitly, let $p(s) = (\log s)/\sqrt{s}$. This function is decreasing for sufficiently large s , and

$$\frac{p(s^2)}{p(s)} = \frac{2}{\sqrt{s}}.$$

In reverse order, the parameters at which reductions are performed start above X and are successively squared. Increasing X therefore makes the sum of $C_2 p(\ell)$ at most $1/2$. The sum of $C_3 (\log \ell)/\ell$ is bounded by a constant depending only on r . Iterating (3.21), and using $\prod_j (1 - a_j) \geq 1 - \sum_j a_j$ for these nonnegative losses, gives

$$T_{\text{final}} \geq \frac{1}{2} T_0 - O_r(M_x). \quad (3.22)$$

At the final stage every neighborhood has weight at most e^X . Every triangle still has a middle vertex. At a middle vertex a , the incident triangle mass, including the factor b , is at most $bq_aq(N(a))^2/2$. Hence (3.1) gives

$$T_{\text{final}} \leq \frac{e^X}{2} \sum_b b \sum_{\substack{a \in V(F_b) \\ a \text{ middle}}} q_aq(N(a)) \leq e^X M_x.$$

This estimate also applies if $\ell_0 \leq X$ and no reduction is needed. Together with (3.22), it proves $T_0 = O_r(M_x)$. $\square$

Completion of the proof of Lemma 2.1. Proposition 2.5 bounds the contribution from triangles all of whose vertices are good by $O_r(M_x)$. The contribution from triangles meeting a bad vertex is, by (2.9), at most $O(\Lambda^2) \sum_i w_i y_i$. Taking $B = B(r)$ sufficiently large and then increasing the lower threshold on Λ gives

$$\sum_i x_i t_i \leq B \sum_i x_i d_i + \Lambda^4 \sum_i w_i y_i,$$

which is the dual inequality (2.7). The separation argument in Section 2 now supplies the finite family of operations and rates in (2.5). The constants are chosen in order: first the multiplier bound A and the local estimate constant C_1 , then the reduction constants and X , and finally B and the threshold on Λ . Every choice depends only on r , as required. $\square$

4 Coloring rounds with dependent entry weights

We now use Lemma 2.1 to extend a partial correspondence coloring while controlling the weights of the entries that remain available. Entropy and edge energy, together with compensated weight changes, are features of Johansson's coloring method; see [6, Sections 2 and 6–8]. Here the adjustment operations act on the graph of list entries. Entries in a single list need not evolve independently; the bound on the number of entries from each part in an operation will supply the concentration needed for a simultaneous update.

Fix $r \geq 4$ and let $B = B(r)$ be the constant in Lemma 2.1. Choose

$$K = 16(B + 2), \quad \gamma = 0.003, \quad C = \frac{100K}{\gamma}, \quad k = \left\lceil \frac{C\Delta}{\log \Delta} \right\rceil. \quad (4.1)$$

We prove Theorem 1.1 with $C_r = C$. Restrict each list to exactly k entries, restricting the matchings accordingly, and continue to write $L(v)$ for the resulting lists. For sufficiently large Δ we have $k \leq \Delta$. Throughout the coloring process, Δ denotes the maximum degree of the original graph. Set

$$w_* = \frac{K}{k}, \quad \Lambda = 20 \log \Delta, \quad \tau = \Delta^{-0.10}, \quad \xi = \Delta^{-0.14}, \quad \zeta = \Delta^{-0.15}. \quad (4.2)$$

The parameter τ sets the activation and adjustment rates, ξ is the permitted error per round, and ζ is a smaller tolerance used in the concentration estimates. All asymptotic estimates below are uniform over the graphs, lists, matchings, and states satisfying the stated assumptions, with r fixed. In particular, a factor $\Delta^{o(1)}$ in an upper bound is at most Δ^ε for every fixed $\varepsilon > 0$ once Δ is sufficiently large in terms of r and ε . The parameters U, P_0 in Lemma 2.1 are $\Delta^{o(1)}$, and $R_0 = O(\log \log \Delta)$, at the value of Λ in (4.2).

Consider a proper partial correspondence coloring, and let G' be the subgraph induced by its uncolored vertices. An entry at an uncolored vertex is *available* if it conflicts with none of the colors already chosen at neighboring vertices. Give each entry $i \in L(v)$, for $v \in V(G')$, a nonnegative weight w_i , requiring every positive-weight entry to be available. Let H be the graph on the positive-weight entries whose edges are the forbidden pairs in the restricted matchings. Its parts are the sets $L(v) \cap V(H)$. The map sending an entry to its host vertex is a graph homomorphism $H \rightarrow G'$. A clique in H has distinct hosts, so H is K_r -free. The matching condition also implies that each entry has at most one neighbor in any part. These are precisely the structural assumptions of Lemma 2.1.

Use s_i, d_i, t_i for the neighborhood, incident-edge, and triangle weights in that lemma, calculated in H . We maintain

$$\Delta^{-3} \leq w_i \leq \Delta^{-0.99}, \quad s_i \leq \Lambda \quad (i \in V(H)). \quad (4.3)$$

For $v \in V(G')$ and $vu \in E(G')$, define

$$\begin{aligned} W_v &= \sum_{i \in L(v)} w_i, \\ D_{vu} &= \sum_{\substack{i \in L(v), l \in L(u) \\ il \in E(H)}} w_i w_l, & D_v &= \sum_{u \in N_{G'}(v)} D_{vu}, \\ F_v &= \sum_{i \in L(v)} w_* f(w_i/w_*) + \gamma \log \Delta (K - W_v) + D_v. \end{aligned} \quad (4.4)$$

The entropy sum includes zero-weight entries, using $f(0) = 1$. Both that sum and D_v are nonnegative, and $D_v = \sum_{i \in L(v) \cap V(H)} d_i \leq \Lambda W_v$. Thus an upper bound on F_v prevents the total weight W_v from becoming too small, while coloring neighbors decreases its edge term.

Initially, color no vertices and give every entry weight w_* . Since $w_* = \Theta(\log \Delta / \Delta)$ and $s_i \leq \Delta w_* \leq (K/C) \log \Delta$, the bounds (4.3) hold. Moreover,

$$W_v = K, \quad F_v = D_v \leq \frac{K^2 \Delta}{k} \leq \frac{K^2}{C} \log \Delta. \quad (4.5)$$

The following proposition is the step we shall iterate.

Proposition 4.1 (One coloring round). *For all sufficiently large Δ depending only on r , suppose that a proper partial correspondence coloring and entry weights satisfy availability, (4.3), and*

$$K/2 \leq W_v \leq 2K \quad (v \in V(G')). \quad (4.6)$$

There is an extension of the partial coloring and a choice of weights on the remaining entries that preserve availability and (4.3), and satisfy

$$W_v^{\text{new}} \leq W_v + \xi, \quad F_v^{\text{new}} \leq F_v + \xi, \quad \frac{\deg_{\text{new}}(v)}{\Delta} \leq (1 - \tau) \frac{\deg_{G'}(v)}{\Delta} + \xi \quad (4.7)$$

for every vertex v that remains uncolored.

4.1 The random update and its expectation

Assume the hypotheses of Proposition 4.1; the case $V(G') = \emptyset$ is immediate. Fix the operations and rates a_j given by Lemma 2.1 on H , with listed supports J_j . We use two families of mutually

independent Bernoulli flags. For each entry $z \in V(H)$ an *activation flag* is set with probability τw_z ; it touches every neighbor of z in H , with designated multiplier zero there. For each operation j its flag is set with probability τa_j and touches J_j . A multiplier operation has its prescribed multipliers $m_i^{(j)}$ on J_j , and a drop has designated multiplier zero on its singleton support. The probabilities are valid for large Δ : the nonempty support and rate bound of Lemma 2.1 give $a_j \leq \Lambda^4$.

For each $i \in V(H)$ put $\bar{w}_i = w_i Y_i$. If exactly one set flag touches i , let Y_i be its designated multiplier; if at least two set flags touch i , let $Y_i = 0$. If no set flag touches i , put

$$Y_i = 1 + \tau \left(s_i + \sum_{j \in \mathcal{M}} a_j (1 - m_i^{(j)}) \right), \quad (4.8)$$

where $\mathcal{M}$ indexes the multiplier operations and their multipliers are extended by 1 off the listed supports. Zero-weight entries stay at zero. The coefficient of τ in (4.8) has absolute value at most $\Lambda + \Lambda^4(1 + U)$. The no-flag multiplier is therefore positive and is $1 + O(\tau \Delta^{o(1)})$.

An entry *activates* when its activation flag is set. An activating entry is *successful* if none of its neighbors in H activates. At every host with a successful entry, choose one such entry as its color, using any fixed order on that list to break ties. Successful entries cannot be adjacent in H , and they were available before the round, so this extends the proper partial coloring. Every entry conflicting with a newly chosen color is touched by its activation flag and hence has provisional weight zero, whether or not other flags also touch it. Consequently, after newly colored vertices are removed, all remaining positive provisional weights are available. The set of hosts receiving a color depends only on activation flags, independently of the choices among successful entries.

Until that removal, a bar on any quantity in (4.4) means that it is calculated with the provisional weights on the entire start-of-round graph G' . We may continue to use $E(H)$ in these sums because zero weights remain zero. The compensation in (4.8) cancels the first-order mean loss from activations and multiplier operations. The following estimate shows how the adjustment lemma controls the remaining cost in the potential.

Lemma 4.2 (Expected changes). *For the random update just defined,*

$$\begin{aligned} \mathbb{E} \bar{W}_v &\leq W_v + o(\xi), \\ \mathbb{E} \bar{F}_v &\leq F_v + \tau(B + 1)D_v + o(\xi), \\ \mathbb{E} \bar{D}_{vu} &\geq \frac{1}{2}D_{vu} \quad (vu \in E(G')). \end{aligned} \quad (4.9)$$

Proof. Let a_i^- be the sum of the rates of drops at i . Expanding according to the set flags touching one or two entries gives the following identities, with $il \in E(H)$ in the last line:

$$\begin{aligned} \mathbb{E}(Y_i - 1) &= -\tau a_i^- + O(\tau^2 \Delta^{o(1)}), \\ \mathbb{E} f(Y_i) &= \tau \left(s_i + a_i^- + \sum_{j \in \mathcal{M}} a_j f(m_i^{(j)}) \right) + O(\tau^2 \Delta^{o(1)}), \\ \mathbb{E}[(Y_i - 1)(Y_l - 1)] &= \tau \sum_{z \in N_H(i) \cap N_H(l)} w_z \\ &\quad + \tau \sum_{j \in \mathcal{M}} a_j (m_i^{(j)} - 1)(m_l^{(j)} - 1) + O(\tau^2 \Delta^{o(1)}). \end{aligned} \quad (4.10)$$

Here is a justification of the errors and the first-order terms. The sum of the probabilities of flags touching either of two fixed entries is at most $2\tau(\Lambda + \Lambda^4)$. Independence bounds both

the probability of two or more such flags and the total error in replacing the exact single-flag probabilities by their marginal probabilities by $O(\tau^2 \Delta^{o(1)})$. All multipliers and their f -values are $\Delta^{o(1)}$, so these errors remain of the stated order. For the first line, the single flags contribute $-s_i - a_i^- + \sum_{j \in \mathcal{M}} a_j (m_i^{(j)} - 1)$ to the coefficient of τ ; (4.8) cancels all but $-a_i^-$. For the second line, use $f(0) = 1$ and $f(1+h) = O(h^2)$ at the no-flag value. For the third line, only flags touching both entries contribute to first order. These are common-neighbor activations and shared multiplier flags. A singleton drop cannot touch both endpoints, and a flag touching only one endpoint has an additional factor $O(\tau \Delta^{o(1)})$ from the other endpoint.

For every $i \in V(H)$, including when $Y_i = 0$, the entropy identity is

$$w_* [f(w_i Y_i / w_*) - f(w_i / w_*)] = w_i \log(w_i / w_*) (Y_i - 1) + w_i f(Y_i). \quad (4.11)$$

Allocate the change in F_v to its positive-weight entries. The contribution of $i \in L(v) \cap V(H)$ is

$$w_i [(\log(w_i / w_*) - \gamma \log \Delta)(Y_i - 1) + f(Y_i)] + \sum_{l \in N_H(i)} w_i w_l (Y_i Y_l - 1). \quad (4.12)$$

Its expectation is at most

$$\tau \left(d_i + t_i + \sum_j a_j g_i^{(j)} \right) + O((w_i + d_i) \tau^2 \Delta^{o(1)}). \quad (4.13)$$

Indeed, (4.10) gives the activation terms d_i from entropy and t_i from edge correlations. For a multiplier operation, the compensated linear terms cancel, leaving exactly $w_i f(m_i^{(j)})$ and the quadratic edge terms that define $g_i^{(j)}$. A drop at i contributes $w_i(1 - \log(w_i / w_*) + \gamma \log \Delta) \leq \Lambda w_i$ to the non-edge part and $-d_i$ to the edge part. A drop at a neighbor l contributes $-w_i w_l$. These are bounded by the drop vector in Lemma 2.1. Finally, $|\log(w_i / w_*)| = O(\log \Delta)$ by (4.3); absorbing these logarithms gives the error in (4.13).

Sum (4.13) over $L(v) \cap V(H)$ and use $t_i + \sum_j a_j g_i^{(j)} \leq B d_i$. Since $D_v \leq \Lambda W_v$ and $W_v \leq 2K$, the total error is $O((W_v + D_v) \tau^2 \Delta^{o(1)}) = o(\xi)$. This proves the potential estimate. The first line of (4.10) proves the mass estimate. For each edge il in the sum defining D_{vu} , the probability that neither endpoint is touched by a set flag is $1 - O(\tau \Delta^{o(1)})$; on this event, $Y_i Y_l = 1 + O(\tau \Delta^{o(1)})$. Nonnegativity on the other events therefore gives $\mathbb{E}(w_i w_l Y_i Y_l) \geq w_i w_l / 2$ for large Δ . Summing proves the last assertion. $\square$

4.2 Concentration with bounded input participation

We next make the expectation bounds simultaneous at all vertices. The relevant concentration principle is a bounded-variable form of the generalized Hölder inequality of Finner [16] and the read- k concentration method of Gavinsky, Lovett, Saks, and Srinivasan [17]. We give the short proof, including the exponential-moment estimate of Hoeffding [18].

Lemma 4.3 (Bounded input participation). *Let $X_1, \dots, X_m$ be bounded real functions of finitely many independent inputs. Suppose that X_j has range of length at most b_j , and that each input is used by at most p of the functions, where $p \geq 1$ is real. If $\sum_j b_j^2 > 0$, then for every $a > 0$,*

$$\mathbb{P} \left(\sum_j (X_j - \mathbb{E} X_j) \geq a \right), \quad \mathbb{P} \left(\sum_j (X_j - \mathbb{E} X_j) \leq -a \right) \leq \exp \left(-\frac{2a^2}{p \sum_j b_j^2} \right). \quad (4.14)$$

If $\sum_j b_j^2 = 0$, both probabilities are zero.

Proof. For nonnegative bounded functions Z_j with the same dependence sets, we first prove

$$\mathbb{E} \prod_j Z_j \leq \prod_j (\mathbb{E} Z_j^p)^{1/p}. \quad (4.15)$$

Induct on the number of independent inputs. Condition on all inputs except the last, and let I index the functions using that last input. Hölder's inequality with exponent p for the $|I| \leq p$ factors, and with a constant factor 1 of reciprocal exponent $1 - |I|/p$ when $|I| < p$, bounds this conditional integral by $\prod_j \tilde{Z}_j$, where $\tilde{Z}_j = (\mathbb{E}_{\text{last}} Z_j^p)^{1/p}$ for $j \in I$ and $\tilde{Z}_j = Z_j$ otherwise. Each $\tilde{Z}_j$ uses only the remaining inputs of Z_j , so the participation bound still holds. The induction hypothesis applies and gives $\mathbb{E} \prod_j \tilde{Z}_j \leq \prod_j (\mathbb{E} \tilde{Z}_j^p)^{1/p} = \prod_j (\mathbb{E} Z_j^p)^{1/p}$, proving (4.15).

If a random variable X has range of length b , then

$$\log \mathbb{E} e^{s(X - \mathbb{E}X)} \leq \frac{s^2 b^2}{8} \quad (s \in \mathbb{R}). \quad (4.16)$$

To see this, the second derivative of the logarithm on the left is the variance under the exponentially tilted law. Any law supported on an interval of length b has variance at most $b^2/4$, by measuring squared distance from the midpoint. The logarithm and its first derivative vanish at zero, which proves (4.16). Apply (4.15) to $Z_j = e^{s(X_j - \mathbb{E}X_j)}$ and then (4.16) with parameter ps . The resulting moment bound is $\exp(ps^2 \sum_j b_j^2/8)$. Exponential Markov with $s = 4a/(p \sum_j b_j^2)$ proves the upper tail; replacing X_j by $-X_j$ proves the lower tail. $\square$

Apply Lemma 4.3 first to $\bar{W}_v$ and to the non-edge part $\bar{F}_v - \bar{D}_v$. Apart from an additive constant, each is a sum of at most $k \leq \Delta$ functions, one for each entry of $L(v)$, of range length at most $\Delta^{-0.99+o(1)}$. This follows from (4.3), (4.8), and $U = \Delta^{o(1)}$; the logarithmic factors in the entropy term are absorbed in $\Delta^{o(1)}$. This also covers arbitrarily small positive provisional weights: with $0 \log 0 = 0$, $\sup_{0 \leq z \leq b} |z \log z| = b \log(1/b)$ for $0 < b < e^{-1}$. A multiplier flag occurs in at most P_0 of these functions, by its support bound in a single part. A drop flag occurs in at most one, and an activation flag occurs in at most one because its activating entry has at most one neighbor in $L(v)$. Thus $p = P_0$ is valid. In particular, each requirement

$$\bar{W}_v \leq \mathbb{E} \bar{W}_v + \zeta, \quad \bar{F}_v - \bar{D}_v \leq \mathbb{E}(\bar{F}_v - \bar{D}_v) + \zeta \quad (4.17)$$

fails with probability at most $\exp(-\Delta^{0.5})$ for large Δ : the exponent in (4.14) is at least $\Delta^{0.68-o(1)}$.

For an edge $vu \in E(G')$, the matching sum $\bar{D}_{vu}$ has at most k terms, each of range length at most $\Delta^{-1.98+o(1)}$. Every entry at either endpoint occurs in at most one term, so each flag occurs in at most $2P_0$ terms. Applying Lemma 4.3 with $p = 2P_0$ shows that

$$|\bar{D}_{vu} - \mathbb{E} \bar{D}_{vu}| \leq \Delta^{-1.25} \quad (4.18)$$

fails with probability at most $2 \exp(-\Delta^{0.4})$. Here the exponent is at least $\Delta^{0.46-o(1)}$. This separation into non-edge terms and individual matching sums is what permits the participation bound to apply; no independence among entries of a list is assumed.

For ordinary list coloring, operations can instead be chosen separately in each graph induced by one common color label. Sampling their flags independently makes contributions of distinct colors independent, so the corresponding concentration bound has $p = 1$. Arbitrary edge matchings provide no such common labels; the per-part support bound replaces that independence here.

4.3 Activations that produce colors

The expectation estimate allows a potential increase of order τD_v . To offset it, we must show that enough neighbors receive colors, with their contributions measured both by D_{vu} and by $1/\Delta$. Fix $v \in V(G')$, and take one of the coefficient families

$$b_u = D_{vu} \quad (u \in N_{G'}(v)), \quad \text{or} \quad b_u = 1/\Delta \quad (u \in N_{G'}(v)). \quad (4.19)$$

In either case $0 \leq b_u \leq 2K\Delta^{-0.99}$: for the first family, the matching property gives $D_{vu} \leq \Delta^{-0.99}W_v$. Let A_u indicate that some entry at u activates. These indicators are independent over the hosts u , and

$$\mathbb{P}(A_u = 1) \geq 1 - e^{-\tau W_u} \geq \tau K/4 \quad (4.20)$$

for large Δ , since $K/2 \leq W_u \leq 2K$. Lemma 4.3 with $p = 1$ and $\sum_u b_u^2 \leq 4K^2\Delta^{-0.98}$ gives

$$\sum_{u \in N_{G'}(v)} b_u A_u \geq \frac{\tau K}{4} \sum_{u \in N_{G'}(v)} b_u - \zeta \quad (4.21)$$

except with probability at most $\exp(-\Delta^{0.5})$. The next lemma bounds the loss caused by conflicting activations.

Lemma 4.4 (Collision bound). *Let $\mathcal{E}_v$ be the set of edges of H with an endpoint hosted at a neighbor of v in G' , and let X_v count the edges of $\mathcal{E}_v$ whose two endpoints activate. Then*

$$\mathbb{P}(4K\Delta^{-0.99}X_v > \zeta) \leq O(\Delta^3)e^{e-1-\Delta^{0.01}} + e^{-\Delta^{0.8}}. \quad (4.22)$$

For either coefficient family in (4.19), the sum of b_u over hosts u counted by A_u that receive no color is at most $4K\Delta^{-0.99}X_v$.

Proof. A host with an activating entry but no successful entry contains an endpoint of an edge counted by X_v . Each such edge has at most two hosts and each coefficient is at most $2K\Delta^{-0.99}$, proving the final assertion. Moreover,

$$\mu_v := \mathbb{E}X_v \leq \tau^2 \sum_{u \in N_{G'}(v)} \sum_{i \in L(u) \cap V(H)} w_i s_i \leq 2K\Delta\tau^2\Lambda = \Delta^{0.80+o(1)}. \quad (4.23)$$

For a possible endpoint i of an edge in $\mathcal{E}_v$, the number of its activating neighbors is a sum of independent indicators of total mean $\tau s_i \leq \tau\Lambda \leq 1$. Its exponential moment at parameter 1 is at most $\exp(e-1)$, so it exceeds $\Delta^{0.01}$ with probability at most $\exp(e-1-\Delta^{0.01})$. All possible endpoints are hosted within distance two of v in G' , and there are at most $k \leq \Delta$ entries at each host. A union bound over $O(\Delta^3)$ endpoints shows that, apart from the first error term in (4.22), the graph formed by the edges counted by X_v has maximum degree at most $\Delta^{0.01}$.

If $4K\Delta^{-0.99}X_v > \zeta$, then $X_v > \Delta^{0.84}/(4K)$. In a graph of maximum degree at most $\Delta^{0.01}$, greedy selection of disjoint edges produces a matching of size at least $X_v/(2\Delta^{0.01})$. For large Δ this is at least $m = \lceil \Delta^{0.82} \rceil$. For $e = il \in \mathcal{E}_v$ write $q_e = \tau^2 w_i w_l$. The events that both endpoints activate are independent along any matching, because its edges have disjoint endpoints and the activation flags are independent. Consequently,

$$\begin{aligned} \mathbb{P} \left(\begin{array}{l} \text{some size-}m \text{ matching in } \mathcal{E}_v \\ \text{has all endpoints activating} \end{array} \right) &\leq \sum_{\substack{M \subseteq \mathcal{E}_v \\ M \text{ matching, } |M|=m}} \prod_{e \in M} q_e \\ &\leq \frac{(\sum_{e \in \mathcal{E}_v} q_e)^m}{m!} = \frac{\mu_v^m}{m!} \leq \left(\frac{e\mu_v}{m} \right)^m \leq e^{-\Delta^{0.8}}. \end{aligned} \quad (4.24)$$

The second inequality follows by expanding the power: each matching appears $m!$ times, and all other terms are nonnegative. The final inequality uses (4.23) and $m = \Delta^{0.82+o(1)}$; it is immediate if $\mu_v = 0$. Combining the two exceptional events proves the lemma. $\square$

Together, (4.21) and Lemma 4.4 imply that, for either choice in (4.19), we may require

$$\sum_{\substack{u \in N_{G'}(v) \\ u \text{ receives a color}}} b_u \geq \frac{\tau K}{4} \sum_{u \in N_{G'}(v)} b_u - 2\zeta. \quad (4.25)$$

Let E_v be the event that (4.17), (4.18) for an edge incident with v , or (4.25) for one of the two coefficient families fails. A union bound gives, for all sufficiently large Δ ,

$$\mathbb{P}(E_v) \leq \exp(-\Delta^{0.005}). \quad (4.26)$$

4.4 Simultaneous bounds and restoration of the weight conditions

We use the Lovász local lemma [14] to avoid every E_v . Assign each multiplier flag to the host of a chosen center of a radius- R_0 ball in H containing its listed support. Assign activation and drop flags to the hosts of the activating and dropped entries, respectively. Every input determining E_v is assigned within distance $R_0 + 3$ of v in G' . Indeed, the host projection does not increase distances; the provisional weights on an incident matching use flags touching entries at its two hosts; and whether a neighbor of v receives a color is determined by activation flags at distance at most two from v . Events at distance greater than $2(R_0 + 3)$ therefore involve disjoint sets of independent inputs. A bound for the maximum degree of a dependency graph is

$$d = (\Delta + 1)^{2R_0+10} \leq \exp(O(\log \Delta \log \log \Delta)). \quad (4.27)$$

For completeness, the symmetric local lemma says that finitely many events of probability at most q , with a dependency graph of maximum degree at most $d \geq 1$, can all be avoided if $q \leq 1/(e(d+1))$. Put $x = 1/(d+1)$, so $q \leq x(1-x)^d$. Induction on the number of conditioned events shows that any event, conditioned on the avoidance of any set of the others, has probability at most x . To make the induction step, split the conditioning set into nonneighbors and at most d neighbors of the event. Independence bounds the numerator after conditioning on the nonneighbors by q , and the induction hypothesis bounds the probability of avoiding the neighbors, under that conditioning, below by $(1-x)^d$. The same induction ensures that all conditioning probabilities are positive. Multiplying these positive conditional avoidance probabilities proves joint avoidance. By (4.26) and (4.27), its hypothesis holds here for large Δ .

Proof of Proposition 4.1. Fix a trial avoiding all E_v . Lemma 4.2, (4.17), and (4.18) summed over the edges incident with v give

$$\bar{W}_v \leq W_v + o(\xi), \quad \bar{F}_v \leq F_v + \tau(B+1)D_v + o(\xi), \quad (4.28)$$

because $\zeta + \Delta\Delta^{-1.25} = o(\xi)$. Remove the vertices colored in this trial. At a remaining vertex v , only the edge part of its potential changes. Each removed neighbor u contributes $\bar{D}_{vu} \geq D_{vu}/2 - \Delta^{-1.25}$ by (4.9) and (4.18). Using (4.25) with $b_u = D_{vu}$, the total decrease is at least

$$\sum_{\substack{u \in N_{G'}(v) \\ u \text{ receives a color}}} \bar{D}_{vu} \geq \frac{\tau K}{8} D_v - \zeta - \Delta^{-0.25}. \quad (4.29)$$

Since $K/8 > B + 1$, the potential after removal is at most $F_v + o(\xi)$. The same requirement with $b_u = 1/\Delta$ gives the degree bound in (4.7), because $K/4 \geq 1$ and $2\zeta = o(\xi)$. It remains to restore (4.3) without losing the mass and potential estimates.

To *trim* an entry means to set its weight to zero. If its current positive weight is z , the change in the non-edge part of its host's potential is exactly

$$c(z) := z(1 - \log(z/w_*) + \gamma \log \Delta). \quad (4.30)$$

First trim all entries with $z > \Delta^{-0.99}$. Their costs are nonpositive for large Δ , since $\log(\Delta^{-0.99}/w_*) = 0.01 \log \Delta - O(\log \log \Delta)$ and $\gamma < 0.01$. Next trim all positive entries with $z < \Delta^{-3}$. The function $c(z)$ is increasing on this interval for large Δ and has limit zero at zero, so each cost is at most $O(\Delta^{-3} \log \Delta)$. There are at most $k \leq \Delta$ entries at a host, giving total cost $o(\xi)$ there. Both trimming operations can only decrease edge terms.

At the resulting weights, mark each positive entry i with $s_i > \Lambda$, and trim all marked entries simultaneously. For a marked entry of weight z , (4.30) is at most Λz on the restored weight interval. To account for the edge loss without double counting, let M be the marked entries and fix a host v . In the next identity, H, w_i, s_i, D_v refer to the remaining graph and weights immediately before this last trim, and D'_v is the edge term after it. Then

$$D_v - D'_v = \sum_{i \in M \cap L(v)} w_i s_i + \sum_{i \in (V(H) \setminus M) \cap L(v)} w_i \sum_{l \in M \cap N_H(i)} w_l \geq \sum_{i \in M \cap L(v)} w_i s_i. \quad (4.31)$$

Each deleted edge in D_v has exactly one endpoint hosted at v , so it occurs in precisely one of these two sums. The first sum pays for all non-edge costs at v , since $w_i s_i > \Lambda w_i$ for marked entries. This remains true when both ends of an edge are trimmed. This last trim does not increase any F_v . The unmarked entries had $s_i \leq \Lambda$ before the trim, and deleting weights only reduces their neighborhood weights, so the remaining entries satisfy (4.3).

All trims preserve availability and do not increase W_v . Combining their costs with (4.28) and (4.29) proves the mass and potential bounds in (4.7) for sufficiently large Δ . $\square$

5 Iteration and completion

We finish the proof of Theorem 1.1 by iterating Proposition 4.1. The potential keeps a fixed positive amount of weight at each remaining vertex, while the maximum remaining degree decreases to a smaller order than the number of available positive-weight entries.

Proof of Theorem 1.1. Start from the empty partial coloring and the weights in (4.5), and set

$$T_* = \left\lceil 2\tau^{-1} \log \Delta \right\rceil. \quad (5.1)$$

Perform T_* rounds, stopping earlier if every vertex receives a color. We verify inductively that the mass assumption (4.6) remains valid. After $j \leq T_*$ completed rounds, every remaining vertex satisfies

$$W_v \leq K + j\xi, \quad \gamma \log \Delta (K - W_v) \leq F_v \leq \frac{K^2}{C} \log \Delta + j\xi. \quad (5.2)$$

The upper bounds follow from (4.5) and (4.7); the lower bound on F_v follows from the nonnegative entropy and edge terms in (4.4). Since $T_*\xi = O(\Delta^{-0.04} \log \Delta) = o(1)$ and $C = 100K/\gamma$, these inequalities give

$$K - \frac{K}{100} - \frac{T_*\xi}{\gamma \log \Delta} \leq W_v \leq K + T_*\xi. \quad (5.3)$$

For sufficiently large Δ , both bounds lie inside $[K/2, 2K]$. Availability and (4.3) are preserved by the proposition, so each successive round is justified.

If vertices remain after T_* rounds, repeated use of the degree inequality in (4.7) gives

$$\frac{\deg_{\text{remaining}}(v)}{\Delta} \leq (1 - \tau)^{T_*} + \xi \sum_{j=0}^{T_*-1} (1 - \tau)^j \leq \Delta^{-2} + \frac{\xi}{\tau} = \Delta^{-2} + \Delta^{-0.04}. \quad (5.4)$$

On the other hand, the mass lower bound and the upper bound on each positive entry imply

$$|\{i \in L(v) : w_i > 0\}| \geq \frac{K}{2} \Delta^{0.99}. \quad (5.5)$$

For large Δ this exceeds the remaining maximum degree, which is at most $\Delta^{-1} + \Delta^{0.96}$ by (5.4). Color the remaining vertices in any order, choosing at each step a positive-weight entry compatible with the colors already chosen. Every such entry is available with respect to the partial coloring from the rounds. Each neighbor colored during this final procedure forbids at most one entry, by the matching condition. Inequality (5.5) therefore guarantees a choice at every step.

The resulting coloring chooses one entry from each restricted list and avoids every forbidden pair, so it is also a coloring for the original lists and matchings. All lower thresholds on Δ in the proof depend only on r . Choosing a single $\Delta_r \geq 3$ above them, together with $C_r = C$ from (4.1), proves Theorem 1.1. $\square$

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